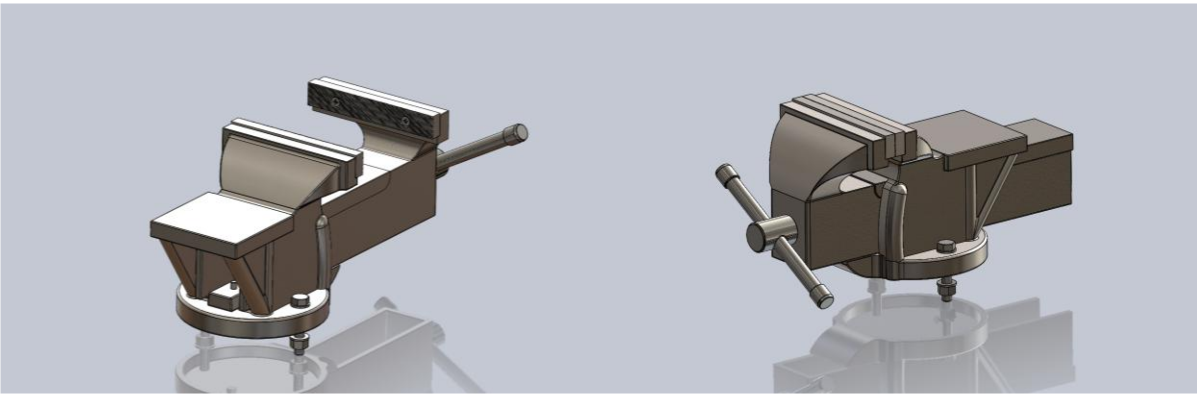
Trinity College Dublin
Coláiste na Tríonóide, Baile Átha Cliath
The University of Dublin

3B7: Manufacturing Technology and Systems

Design for Manufacture and Assembly (DFMA) Assignment



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Introduction

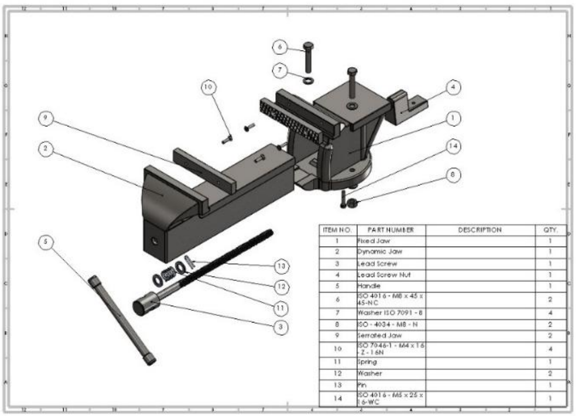
The product we chose to analyse is a table-mounted vice, a standard workshop tool designed to hold objects securely for tasks such as cutting, filing, drilling or shaping. The vice has six distinct components, including a static jaw, dynamic jaw, lead screw, handle, lead screw nut and serrated jaws. The jaws we are using are made from robust cast iron, which can withstand significant compressive forces, while the lead screw and handle are steel to deliver the torque necessary for clamping. Vices are used in both industrial and DIY environments, where secure clamping under repetitive, high-load conditions is essential. This causes them to endure frequent tightening and loosening cycles, making material selection and manufacturing processes extremely important.

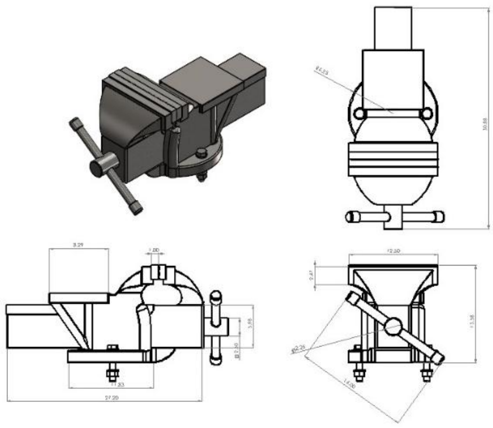
Our table-mounted vice was chosen specifically because it clearly meets the DFMA criteria. It contains at least five distinct metal components (excluding standard fasteners) and is constructed primarily from robust metals like cast iron and steel. It is easily disassembled with simple hand tools, making it ideal for reverse engineering and detailed SolidWorks modelling. The vice's design also provides excellent opportunities for DFMA improvements, including enhancements to jaw design, streamlined manufacturing processes, and overall cost reductions, making it an ideal subject for our analysis.

Given the vice's widespread application in industrial and DIY environments, this report will explore the DFMA principles to evaluate the current design, propose practical improvements, and validate these modifications using detailed SolidWorks models and analyses.

Component Analysis

There are 6 key components to our table vice, the table below shows an overview of their material and manufacturing techniques





Component	Material	Manufacturing Technique
Fixed Jaw	Cast Iron	Casting, machining and drilling
Dynamic Jaw	Cast Iron	Casting, machining and drilling
Lead Screw	Hardened Steel	Thread Rolling /Whirling
Handle	Mild Steel	Rolling
1 x Lead Screw Nut	Hardened Steel	Rolling and Machining
2 x Serrated Jaw	Hardened Steel	Machining

Static and Dynamic Jaws

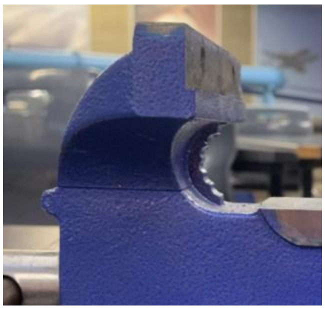
Function – Static Jaw

The primary purpose is to be a stationary element attached to the work bend, allowing the dynamic jaw to clamp an object in place securely. The jaw is secured to the bench using two bolts.

Function – Dynamic Jaw

The dynamic jaw is positioned opposite the fixed jaw. When the handle is rotated, it rotates the lead screw, which in turn moves the dynamic jaw either forward or backwards, creating a compressive force against the static jaw.

Material



We determined the material to be cast metal from the casting lines on the vice as seen in the image on the right. This led us to conduct research and discover that the jaws are made out of cast iron. It's impossible to definitively say what type of cast iron, but our research suggests it is likely to be Grade 250, a composition often used in workshop tools. Cast iron is an excellent material to use due to its cost-effectiveness, exceptional compressive strength, easy machinability, and fatigue resistance (Shane, 2025).

Manufacturing Process

The jaws are produced through a casting process; we discovered this from the casting lines and complex geometry of the vice. Casting involves pouring molten cast iron into a mould cavity designed to replicate the final shape of the jaws. The molten iron then cools and solidifies within the mould, transferring heat to the mould walls, eventually forming the desired geometry of the part and the mould is broken away (Science Direct, 2003). The surfaces of the jaws are then machined and finished; this involves milling, a machining process that uses rotating cutters to remove material from a workpiece to achieve precise dimensions and surface finishes (Ye, 2023). Holes are then drilled and tapped into both jaws. This requires proper alignment to ensure all five holes on the static vice and two on the dynamic vice are correctly placed. The Jaws are then prepped to be painted; this requires removing any grease or dirt first to ensure proper adhesion. The paint acts as a barrier to erosion.

Lead Screw

Function

The lead screw is the core mechanical element that drives the dynamic jaw. As the user turns the handle, the screw rotates within a threaded nut, converting rotational motion into linear displacement. This allows the dynamic jaw to move forward or backwards, allowing the vice to clamp or release objects. A compression spring is housed behind the handle to maintain tension and ensure the handle returns smoothly when released, preventing backlash and helping maintain consistent pressure during use (Industrial Quick Search, N/A).

Material

We were unable to determine the specific material of our lead screw; this is because we were unable to remove it from the dynamic vice but we found that they are made out of carbon steel, stainless steel, aluminium, and titanium, depending on their application (Industrial Quick Search, N/A). We believe the optimal material for our lead screw would be pre-hardened medium carbon steel (4140). This steel typically provides high tensile strength 850 - 1000 MPa, excellent fatigue resistance, and good toughness, making it well suited for parts subjected to constant torque and axial loads (Rebecca Piccoli, 2025).

Manufacturing Technique
Before the thread can be machined, a raw steel bar needs to be straightened and cut to length. The lead screw has an ACME thread, this could have been manufactured using 2 techniques, thread rolling or thread whirling.

Thread Rolling: A pair of profiled tools simultaneously revolve around the steel rod, using cold forming to press the thread shape into the surface. This results in a work-hardened, smooth, and strong thread. This can be done using large production volumes (Moser, 2020).

Thread Whirling: Involves a ring-shaped cutter that rotates around the stationary rod, cutting the thread profile rather than forming it, this produces a very precise thread with small tolerance. This is generally more expensive and requires smaller production volumes. (Moser, 2020)

While both processes could be used, we believe thread rolling was most likely used to manufacture our lead screw because it is cheaper, provides better surface finish and longer service life.

Handle

Function:
The handle is the user-operated lever that rotates the lead screw and causes the dynamic jaw to move forward or backwards. It provides the mechanical advantage needed to apply the clamping force. It must be strong enough to withstand high torque yet designed to fail by bending, acting as a warning to prevent you from damaging the vice (Grays Tools, 2016).

Material:
Based on the function of the handle and an approximation of the maximum clamping force, we were able to accurately calculate that the handle is made out of mild steel. We calculated the maximum torque applied during operation and the resulting shear stress experienced by the handle. This correlated to a shear yield strength of approximately 145 MPa, which is consistent with the known properties of mild steel.

Manufacturing Technique:
The process begins by cutting a metal rod to the desired length and machining it on a lathe to achieve the correct diameter. The ends are turned down and fitted with spherical knobs, which are secured by welding. These knobs provide a comfortable grip and prevent the handle from sliding out during use (Wilton Vice Parts, 2017) . After assembly, the handle was polished to remove sharp edges and improve aesthetics (Fireball Tool, 2025).

Serrated Jaws

Function:
The serrated jaws are replaceable inserts mounted to the clamping faces of both the static and dynamic jaws. Their primary role is to securely grip the workpiece by embedding sharp, patterned teeth into its surface, which concentrates the clamping force, enhances grip, and prevents slippage. They are replaceable, extending the lifespan of the overall vice (Grays Tools, 2016).

Material:

Based on our understanding of the table vice, we established that the jaws were made of an extremely tough material such as hardened steel. After more research, we discovered they are often made of A2 tool hardened steel (Inception Machines, 2025). This material provides a good balance between hardness, toughness, and wear resistance when properly treated.

Manufacturing Technique:

The jaw is initially cut from bar stock and then machined into shape. The serrated pattern is typically created by milling. This crosshatch pattern results in sharp, pyramidal teeth that enhance the jaw's gripping ability. (Wilson, 2017). Following machining, the jaws undergo a carefully controlled heat treatment to harden the steel, significantly improving their durability and wear resistance. The process starts with slow heating, at a rate not exceeding 222°C per hour, until reaching temperatures between 941-954°C. After holding the jaws at this temperature for at least two hours to ensure uniform heating, they are quenched by air cooling. To reduce brittleness from quenching, tempering is performed by reheating the jaws to 190-220°C, followed by controlled cooling. Finally, the jaws are ground to achieve a smooth, flat surface, ensuring they fit precisely into the vice. (Unionfab, 2025)

Lead Screw Nut

Function:

The nut is the threaded component that mates with the lead screw. It is fixed inside the static jaw by a hex bolt and stays stationary as the lead screw turns within it. This allows rotational motion from the handle to be converted into the linear movement of the dynamic jaw, enabling clamping.

Material Technique:

The nut within our jaw is made from hardened steel and the reason for this is because of its high tensile and shear strength, which makes it capable of withstanding the repeated axial loads applied during clamping. It very resistant to wear, which helps keep the threads in good shape even after a lot of use. This means the nut won't easily deform or wear down, even with regular tightening and heavy loads.

Manufacturing Techniques:

The nut is turned from a steel rod and then a drilled and tapped, with two holes, one to secure to the static jaw and a second hole machined with an ACME thread for the lead screw to go through. A heat treatment, most likely quenching and temping is then done to the part, this is to increase wear resistance and load capacity.

Hex Screws, Washers, Nuts and Spring

The screws, washers and nuts are all key component in assembling our vice. The vice uses standardised components that would be purchased, and used for assembly, these components consist of:

- M4 Countersink Screws x 4 – connect the serrated jaws to the two jaw faces
- M8 Hex Bolts x 2 – secure the fixed jaw to the table
- M8 Washers ISO7091 x 4
- M8 Hex Nuts ISO4034 x 2
- M5 Hex Bolt ISO4016 x1
- M5 Hex Nut ISO4016 x 1
- Compression Spring x 1

Assembly Analysis

In order to determine the optimal assembly process, the vice was disassembled and reassembled multiple times. We carried out the process in order to find a method that was efficient but resulted in a well assembled device. After carrying it out multiple times, we found that by assembling all the small components together first, and then assembling the larger pieces, we ended up with a completed device that tightly assembled in a short amount of time.

Firstly, the fixed jaw and lead screw nut were screwed together. Then the 2 serrated jaws were screwed to either side of the fixed jaw and the dynamic jaw using 2 screws each. The fixed jaw was then screwed onto the table using 2 screws, 2 bolts and four washers. A washer was placed on either screw, then the screws were passed through a hole in the fixed jaw and the table. A second washer was fitted onto each screw. The screws were fastened to the table and the jaw using 2 bolts on the underside of the table. Finally, the dynamic jaw was fitted to the fixed jaw by passing the bolt through the lead screw nut connected to the fixed jaw. The handle was spun so that the two jaws were positioned close together.

It is worth noting that certain parts of the assembly could not be taken apart. The lead screw within the dynamic jaw could not be taken out as the mechanism was spring loaded. If we took the spring out, we were afraid that we would not be able to get it back in and the whole system would no longer work. The end pieces of the handle were welded on, meaning that there was no way to take the handle out of the lead screw. For these processes we estimated the time that would be needed to carry out each one.

Assembly Time Calculation

To measure the assembly time of the device we used the stopwatch time study method. This is a direct measurement process which allows us to accurately measure the assembly time of the device. This allowed us to accurately see how long it took to assemble the device. We recorded each assembly stage and timed how long each one took.

Task	Time
Attaching the Lead Screw Nut to the Fixed Jaw	27 seconds
Screwing Serrated Jaw onto the Fixed Jaw	22.8 seconds
Screwing Serrated Jaw onto the Dynamic Jaw	25.1 seconds
Assembly of the Spring onto the Lead Screw	30 seconds (estimate)
Assembly of the Lead Screw and Dynamic Jaw	30 seconds (estimate)
Assembly of the Handle Through the Lead Screw	5 minutes (estimate)
Attaching the Fixed Jaw to the Table	30.2 seconds
Screwing the Dynamic Jaw into the Lead Screw Nut	47.1 seconds

In total, combining the estimated times with the actual recorded times, the time to assemble the vice is approximately 8 minutes and 32 seconds. There is of course a margin of error here. The approximations for the spring assembly and the handle assembly may be inaccurate. The welding process depends on many different factors. We were unsure how to calculate the actual process time, so many sources were sited to find an estimated time. Welding time depends on factors like, the welding process used (MIG, TIG, Stick, etc.), the size and thickness of the pieces, access, fit-up and welder skill level, and also including the weld prep and cleanup time. It is likely done through TIG welding as it is a more precise and tidy process for smaller parts. The average weld time is around 10-15 seconds per cm for steel

(AWS (American Welding Society), TIG Guidelines). As there are two small parts of circumference 4.7cm, we can say that the total weld time is approx. 2 min 20 seconds. Accounting for some prep time, we can say that the process could be around 5 minutes, but probably less if manufactured in an assembly line.

The spring-lead screw and dynamic jaw-lead screw times were estimated by approximation. As these are simple processes that involve just passing one item through the other, a pretty accurate estimate can be made.

Also, even having carried out the assembly multiple times and taking an average time for each component, the process is carried out by a human after all. This means that the assembly time is most likely going to be inconsistent if the device is put together multiple times. The only way to ensure a higher consistency and reduce this possible error, would be to introduce a robotic assembly process. However, this could prove to be quite costly. Also, this particular vice is intended for personal and small-scale commercial use, so robotic assembly would not be a practical solution.

Detailed Assembly Instructions

For assembly instructions please see the below hyperlinks. These provide access to view the assembly instructions document and the assembly instructions video:

- Assembly Instructions Document
- Assembly Instructions Video

DFMA Improvement

Proposed Modification 1

Remove One of the Screws from the Serrated Jaw:

The current serrated jaw design uses two screws for attachment but by removing one of the screws, this would cut in half all the holes that have to be drilled and tapped since only one screw will be required per jaw. Each hole is time-consuming and expensive because it involves running through a rigorous procedure comprising drilling and threading operations. The single screw will be positioned at the centre of the jaw, and it will be able to maintain its balance due to the support provided by the small platform beneath it, which prevents it from rotating off its axis and is already apparent in the original design. Additionally, the force applied by the screw itself keeps the jaw securely in place, preventing any displacement or unevenness during use, making it an effective alternative to the traditional one screw. A hole per jaw can save costly machining time, tool wear, and labour because machining operations consume a large part of production costs. A small reduction in machining needs easily corresponds to substantial savings when multiplied by large-batch production.

Bench Vices are often produced in large batches due to their widespread use, so even small reductions in machining needs can lead to substantial savings. For examples, if a factory produces 10000 vices annually, halving the holes from two to one per jaw could save hundreds of machining hours and in turn, significantly reduce operational costs. It's estimated that for a reduction in one threaded hole, we can save 42 cent per jaw and save 56 seconds on threading and drilling. For the example factory above this could save approximately 311 hours per year and result in a labour cost reduction of 6,222 euro per year for a worker on a wage of €20 per hour. (CSO, n.d.) It also makes assembly and disassembly easier because it speeds installation or replacement of the parts even more. The fewer screws used, and holes drilled during manufacturing mean fewer steps in the assembly line, thus more rapid production timing and lower labour costs.

Standardising screw types across several models of vices would even further reduce costs because when different models use the same screw type, it makes inventory management a lot simpler, reducing the need to source, store, and track many different fasteners. Not only is inventory cost lowered, but assembly process is simplified because now the workers need to handle and install one and only screw type making it easy to commit less errors or mismatch parts (optimas, 2025).

Proposed Modification 2

Removing the Anvil and its Supports:

The anvil on the bench vice is a flat, hardened and polished surface typically used for drilling, punching and light hammering tasks. It's meant to provide a solid, stable base for shaping or flattening metal without damaging the workbench. However, most users don’t actually use the anvil for these purposes, especially when the vice is mounted to a proper workbench with other tools available. It's meant to provide a solid, stable base for shaping or flattening metal without damaging the workbench but it’s not a replacement for a blacksmiths anvil and should only be used for lighter tasks. Keeping the anvil adds weight, material costs, and unnecessary machining steps which could all me minimised.

By removing the anvil, a large portion of the machining and assembly process can be reduced by a lot. Anvils are typically integrated into the vice via welding or casting, both of which require precise alignment and additional labour. Removing the anvil also provides the opportunity to remove the supports that are connected to it, thus further reducing material cost. Removing these supports also effectively reduces the total machining requirements, allowing for a more streamlined manufacturing process.

Another process that is removed by getting rid of the anvil is that we no longer need to polish the surface and use additional oil coating once the cast has been completed. This process requires workers to spend excessive time degreasing, grinding, and coating, all of which involve abrasive tools and additional manpower, resulting in high expenses and longer assembly time (First Part, 2022).It takes professionals to complete this process 15-30 minutes at a rate of 20 euros per hour resulting in an approximate cost of 3-5 euros per part based off of polishing alone.

Grey cast iron isn’t the most expensive by weight, but it comes to around 90 cent per kg. The weight of the anvil for the specific vice we modelled is approximately 1.4 kg. This allows us to save 1.35 euros per part based on material cost alone which is a small value for one part but when these vices are produced in bulk it will significantly reduce overall costs lead to more additional material than can then be resupplied to the casting process (exxelmet, 2022).

This lighter mass will also allow for a shorter solidification time as part of the casting process which can further decrease the assembly time and improve production. Moreover, lighter castings are also less prone to internal defects since it's easier to manage the mould filling and cooling.

The time saved through reducing the mass of metal being casted can be quantified using Chvorinov's Rule:

Old Volume = $V_{old} = 587.46\ cm^3$ Old Surface Area = $A_{old} = 1071.92\ cm^3$
New Volume = $V_{new} = 394.58\ cm^3$ New Surface Area = $A_{new} = 773.47\ cm^3$

Chvorinov's Rule: $T_s = C\left(\frac{V}{A}\right)^n$, we can say $n = 2$

$$\rightarrow \frac{T_{s,new}}{T_{s,old}} = \frac{C\left(\frac{V_{new}}{A_{new}}\right)^2}{C\left(\frac{V_{old}}{A_{old}}\right)^2} = \frac{\left(\frac{394.58}{773.47}\right)^2}{\left(\frac{587.46}{1071.92}\right)^2} = 0.866 \rightarrow Time\ reduction = 1 - 0.866 = 0.134$$

Therefore, through reducing the mass of the fixed jaw part, we can achieve a 13.4% reduction in casting time. For a factory that mass produces this produce, this would have a very significant reduction in labour, material, and energy costs.

Proposed Modification 3

Using welded rings instead rounded balls at the ends of the handle.

The current design incorporates metal balls at the end of the handle which weigh more than they need to and could be improved by introducing rings to either end of the handle instead. These smaller and lighter rings will be welded on and offer a more efficient and cost-effective solution as opposed to the traditional method. While the rings weigh less, they still contain the necessary strength to prevent them from fracturing. Additionally, since rings can be manufactured from smaller-diameter rods or even stamped from sheet metal, the amount of material required is reduced. Using rings instead of solid balls offers the advantage of reduced weight and while this reduction may appear negligible for a single handle, it becomes a greater factor when applied across large-scale production.

Conclusion

By implementing the above changes, there are certain manufacturing aspects that we have optimised. We have been able to identify a significant weight difference between the original vice design and the modified one. By removing the square anvil and its supports, one of the holes in either serrated jaw, and changing the design of the end-pieces of the handle, we were able to reduce the overall weight of the vice, and in reduce the number of components to be a manufactured.

After evaluating each design on SolidWorks, the original vice design came in at a weight of 8.3kg. The redesigned vice came in at a total weight of 6.9kg. Thus, through our improvements, we managed to save 1.4kg. This reduction in mass is results in a reduced casting time, and overall manufacturing time. This also results in a reduced energy consumption and labour hours, which reduces the cost.

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